



The Zoo Labs Impact Thesis: Measurable, Verifiable, Scalable Conservation Finance

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Abstract

Zoo Labs Foundation is a 501(c)(3) nonprofit research network that operates as the mission layer of the Lux-Hanzo-Zoo ecosystem. This paper formalizes the foundation’s impact thesis: that blockchain infrastructure can transform conservation finance from a trust-dependent, opaque, and unverifiable system into one where every dollar is traceable, every outcome is measured, and every claim is settled against independent evidence. We define an investment-grade impact framework grounded in the IRIS+ standard, the IUCN Red List species prioritization, and the Kunming-Montreal Global Biodiversity Framework targets. Three impact pathways are specified—species protection, habitat conservation, and community partnership—each with measurable indicators, data collection protocols, and target milestones for 2025–2030. We describe the theory of change that connects on-chain infrastructure (DonationRouter, ImpactVoucher, ImpactOracle, TreasuryMultisig, ConservationFeeSplitter) to real-world conservation outcomes, and explain how the sub-DAO structure enables the foundation to serve as a fiscal sponsor and governance umbrella for 35+ aligned conservation DAOs. The thesis demonstrates that a 501(c)(3) research network can operate simultaneously as an on-chain fund issuer, an impact measurement authority, and an ethical anchor for a broader technology ecosystem—without compromising the nonprofit obligations of fiduciary care, restricted-fund compliance, and transparent reporting.

Keywords: impact thesis, conservation finance, theory of change, 501(c)(3), sub-DAO, IRIS+

1 Introduction

The biodiversity crisis is existential. One million species face extinction. Seventy-five percent of terrestrial environments have been significantly altered. The annual funding gap for global conservation stands at approximately \$700 billion [1]. These facts are not disputed by any serious scientific body.

What is disputed is whether the existing institutional infrastructure for conservation finance is adequate to the scale of the crisis. We argue that it is not—not because of insufficient goodwill, but because of structural deficiencies in how conservation capital is raised, allocated, measured, and reported.

Zoo Labs Foundation exists to replace these deficiencies with programmable alternatives. The foundation’s thesis is simple: *Technology and capital can serve the flourishing of all life—but only if anchored by mission, guided by ethics, and accountable to communities and ecosystems.*

This paper formalizes that thesis into an investment-grade impact framework. “Investment-grade” means: quantified targets, standardized metrics, independent verification, and auditable reporting. Conservation donors—whether individual contributors, institutional funders, or governance-token holders—deserve the same rigor that financial investors demand from the companies they fund.

1.1 Position in the Ecosystem

Zoo Labs Foundation occupies a specific position in a three-layer ecosystem:

Lux provides the settlement layer. Hanzo provides AI compute for species identification, habitat monitoring, and threat detection. Zoo provides the mission: conservation outcomes that give the ecosystem purpose and credibility. Sub-DAOs provide decentralized governance for specific conservation domains.

Layer	Entity	Role	Zoo Contribution
Finance	Lux Network	Consensus & settlement	Mission integrity
Compute	Hanzo AI	AI infrastructure	Ethics guidance
Mission	Zoo Labs Foundation	Conservation outcomes	Ethical anchor
Governance	Sub-DAOs	Community ownership	Decentralized execution

Table 1: Ecosystem layers and Zoo’s role.

Barrier	Traditional Symptom	Scale
Trust deficit	Self-reported outcomes	77% of orgs
Overhead opacity	Aggregated financials	63% of orgs
Measurement lag	Annual reports	12–18 month lag
Access friction	\$25+ minimum donation	Excludes micro-donors
Geographic friction	Wire transfer costs	3–8% of amount
Impact uncertainty	No verification	70% of projects

Table 2: Structural barriers in conservation philanthropy.

1.2 Contributions

1. **Impact framework:** A structured theory of change connecting on-chain infrastructure to conservation outcomes, with IRIS+-aligned metrics (Section 3).
2. **Three impact pathways:** Species protection, habitat conservation, and community partnership, each with 2025–2030 targets (Section 4).
3. **Sub-DAO umbrella model:** How a 501(c)(3) serves as fiscal sponsor for 35+ conservation DAOs while maintaining legal compliance (Section 6).
4. **On-chain fund issuer architecture:** How the foundation issues conservation funding instruments (DonationRouter, ImpactVoucher) while meeting nonprofit regulatory requirements (Section 7).
5. **Leverage analysis:** Quantification of how each dollar of direct conservation spending generates >3x indirect impact through ecosystem multiplier effects (Section 8).

2 The Problem

2.1 The Conservation Finance Gap

Global biodiversity conservation requires an estimated \$598–824 billion annually [1]. Current spending totals approximately \$124 billion. The gap is not a matter of awareness—the Kunming-Montreal Global Biodiversity Framework (GBF), adopted in December 2022 by 196 nations, commits to mobilizing at least \$200 billion per year by 2030 [2]. The gap is structural.

2.2 The Opportunity

Blockchain infrastructure solves each barrier:

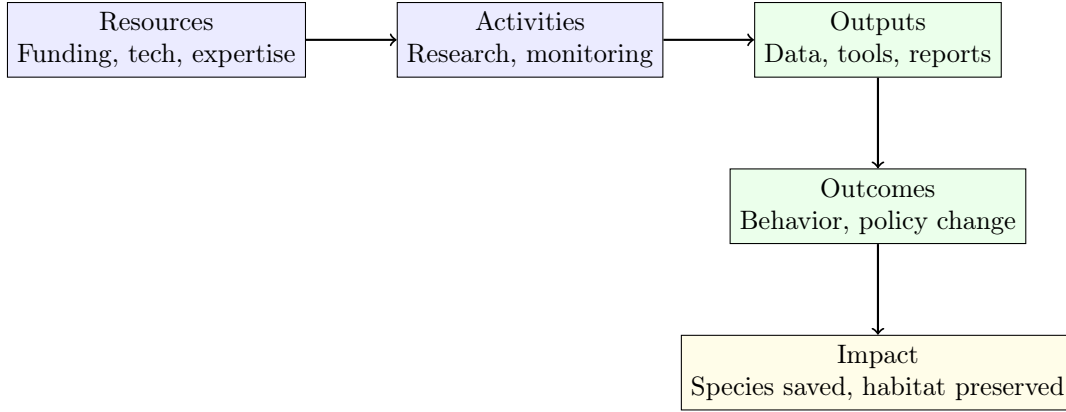


Figure 1: Theory of change: resources $\rightarrow$ activities $\rightarrow$ outputs $\rightarrow$ outcomes $\rightarrow$ impact.

Trust Replace self-reported outcomes with oracle-verified impact metrics.

Overhead Replace manual accounting with smart-contract fund segregation.

Measurement Replace annual reports with continuous on-chain reporting.

Access Replace minimums with frictionless micro-donations ($< \$0.01$ gas cost).

Geography Replace wire transfers with on-chain transfers (0.01% cost).

Verification Replace claims with ImpactVoucher settlement against evidence.

3 Impact Framework

3.1 Theory of Change

At each stage, Zoo’s on-chain infrastructure provides measurement:

- **Resources:** DonationRouter tracks every dollar from donor to program.
- **Activities:** TreasuryMultisig records every disbursement with memo and signer approvals.
- **Outputs:** ImpactVouchers record expected outputs at disbursement time.
- **Outcomes:** ImpactOracle publishes verified outcome metrics from satellite, camera-trap, and sensor data.
- **Impact:** Composite impact score settles ImpactVouchers, closing the feedback loop.

3.2 IRIS+ Alignment

The Global Impact Investing Network’s IRIS+ system provides the standard taxonomy for impact measurement [3]. Zoo metrics map to IRIS+ as follows:

Zoo Metric	IRIS+ Code	Description
Species monitored	OD9535	Org. biodiversity targets
Habitat protected (ha)	OI9803	Land conserved or restored
Community engagement	OI8869	Stakeholders engaged
Restoration (ha)	PI1360	Land improved
Jobs created	PI4427	Employment generated

Table 3: IRIS+ metric alignment.

GBF Target	Zoo Program	Metric
Target 3 (30x30)	Habitat Protection	Hectares protected
Target 4 (Species recovery)	Wildlife Rescue	Population trends
Target 8 (Climate adaptation)	Climate Action	Carbon sequestered
Target 21 (Knowledge)	All programs	Publications
Target 22 (Participation)	Community Conserv.	Communities engaged

Table 4: Alignment with Kunming-Montreal GBF targets.

3.3 IUCN Red List Prioritization

Species-protection efforts are prioritized by IUCN Red List status [4]:

1. Critically Endangered (CR): Highest priority, monitored monthly.
2. Endangered (EN): High priority, monitored quarterly.
3. Vulnerable (VU): Moderate priority, monitored biannually.
4. Near Threatened (NT): Monitored annually.
5. Data Deficient (DD): Targeted for baseline surveys.

This prioritization determines how the DonationRouter’s default allocation weights are set for species-protection programs.

3.4 Kunming-Montreal GBF Alignment

The GBF sets 23 targets for 2030 [2]. Zoo’s programs contribute directly to:

4 Three Impact Pathways

4.1 Pathway 1: Species Protection

Theory of change: Better monitoring → earlier threat detection → faster response → species recovery.

Mechanisms: AI-powered species identification via Hanzo AI models. Camera-trap networks at known wildlife corridors. Anti-poaching technology including real-time GPS tracking of patrol routes. Population genetics through eDNA and genomic sampling.

On-chain integration: Camera-trap data feeds the ImpactOracle’s IMP-001 (Species Population) metric. Anti-poaching patrol GPS data feeds IMP-007 (Patrol Coverage). Both are published on-chain and settle ImpactVouchers for wildlife programs.

Indicator	2025	2027	2030
Priority species monitored	25	50	100
Species with improved status	5	15	30
AI detection accuracy	>90%	>95%	>98%
Peer-reviewed publications	10	50	100

Table 5: Species protection targets.

Indicator	2025	2027	2030
Area monitored	1M ha	5M ha	10M ha
Enhanced protection	100K ha	500K ha	1M ha
Restoration in progress	10K ha	50K ha	100K ha
Deforestation alert response	<72hr	<48hr	<24hr

Table 6: Habitat conservation targets.

4.2 Pathway 2: Habitat Conservation

Theory of change: Better monitoring → faster response → more habitat protected → ecosystem resilience.

Mechanisms: Satellite-based deforestation alerts (MODIS, Landsat, Sentinel-2). Remote-sensing habitat quality assessment. Corridor identification and connectivity analysis. Site-specific restoration planning and monitoring.

On-chain integration: Satellite data feeds IMP-002 (Habitat Protected) and IMP-003 (Deforestation Prevented). Restoration monitoring feeds IMP-006 (Carbon Sequestered). All published via the ImpactOracle.

4.3 Pathway 3: Community Partnership

Theory of change: Genuine partnership → community-led conservation → sustainable outcomes → long-term success.

Principles: Full Free, Prior, and Informed Consent (FPIC) for all activities affecting indigenous and local communities. Communities as rights-holders and co-governors, not beneficiaries. Traditional knowledge protected under community data sovereignty. Fair benefit sharing through on-chain transparent distribution.

On-chain integration: Community engagement data feeds IMP-008 (Community Engagement). Benefit-sharing disbursements are tracked through the TreasuryMultisig with memos identifying community recipients.

5 Five Conservation Principles

From ZIP-500, applied throughout all Zoo operations:

1. **Life-First:** All decisions consider impact on all life, not exclusively human interests. Conservation for ecosystems, not solely for human utility.
2. **Community Partnership:** FPIC, benefit sharing, co-governance. Communities as stewards, not subjects.

Indicator	2025	2027	2030
Communities engaged	50	200	500
FPIC processes completed	100%	100%	100%
Community benefit shared	\$500K	\$2M	\$10M
Community-led initiatives	10	50	200

Table 7: Community partnership targets.

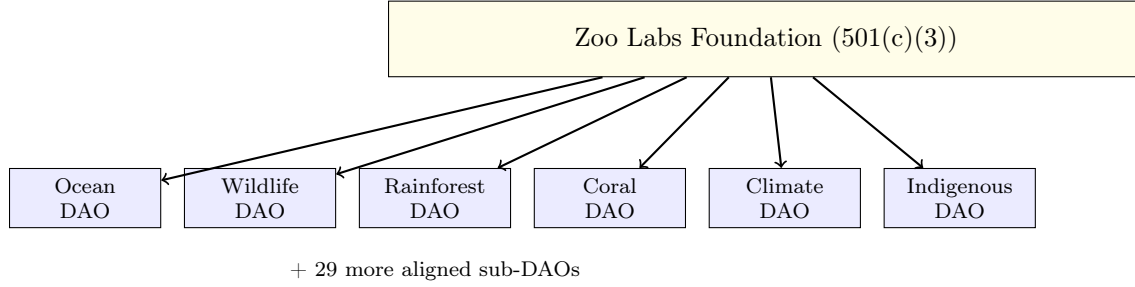


Figure 2: Sub-DAO umbrella structure. Zoo Foundation provides 501(c)(3) fiscal sponsorship and governance guardrails.

3. **Scientific Rigor:** Evidence-based approaches. Peer-reviewed methods. Transparent data.
4. **Long-Term Stewardship:** Durable outcomes, not exit-focused projects. Protocol-level commitments (immutable conservation floors).
5. **Non-Extraction:** No exploitation of communities, traditional knowledge, or ecosystems. Data sovereignty respected.

These principles are not aspirational statements. They are operational constraints enforced through governance votes, smart-contract guards, and sub-DAO membership requirements.

6 Sub-DAO Umbrella Model

6.1 Structure

Zoo Labs Foundation serves as a fiscal sponsor and governance umbrella for aligned conservation DAOs. Sub-DAOs operate under Zoo’s 501(c)(3) status when they meet three requirements:

1. **Mission alignment:** Activities must serve life-first, non-extractive conservation goals.
2. **Governance control:** Zoo Foundation maintains $\geq 51\%$ governance weight (directly or through parent-DAO freeze mechanisms).
3. **Transparency:** Public reporting, open governance, on-chain treasury.

Benefit	Mechanism
501(c)(3) fiscal sponsorship	Tax-deductible donations
Mission alignment framework	ZIP-500 principles
Shared infrastructure	On-chain tools, governance
Network access	Hanzo AI, Lux settlement
Governance support	LuxDAO contracts, voting

Table 8: Sub-DAO benefits from Zoo umbrella.

6.2 What Sub-DAOs Receive

6.3 Planned Sub-DAOs

ZIP-0570 specifies 35 initial sub-DAOs across four categories:

Ecosystem Conservation (10): Ocean, Rainforest, Coral, Arctic, Wetland, Mangrove, Grassland, Mountain, River, Desert.

Species Protection (10): Wildlife, Big Cat, Primate, Elephant, Rhino, Pangolin, Marine Mammal, Bird, Pollinator, Plant.

Mission & Governance (10): Climate, Indigenous, Rewilding, Anti-Poaching, Research, Education, Youth, Carbon, Biodiversity, Seed.

Life Science (5): Longevity, IMAC (ocean DNA sequencing), Genomics, Microbiome, eDNA.

Each sub-DAO deploys a Gnosis Safe treasury managed via LuxDAO governance contracts (ModuleAzoriusV1 for proposal execution, FreezeGuardV1 for parent-DAO override capability).

7 On-Chain Fund Issuer Architecture

Zoo Labs Foundation operates as an on-chain fund issuer: it receives conservation capital (donations, validator-reward splits), allocates it to programs, and produces verifiable impact receipts. This role is functionally similar to a grant-making foundation, but implemented entirely on-chain.

7.1 Regulatory Compliance

As a 501(c)(3), the foundation must satisfy:

- **Public support test:** At least one-third of revenue from public donations or government grants (vs. investment income). On-chain donation records provide auditable evidence.
- **Restricted-fund compliance:** Donor-restricted gifts spent only for designated purposes. Smart-contract fund segregation (Conservation Fund, Grants Fund) enforces this at the protocol level.
- **Form 990 reporting:** Annual disclosure of revenue, expenses, assets, and governance. On-chain data provides the source material; automated quarterly reports reduce manual effort.
- **No private inurement:** No foundation assets benefit insiders. Multi-sig controls and public disbursement records prevent insider self-dealing.

Impact Type	Example	Multiplier
Direct	1 hectare protected	1.0x
Indirect (carbon)	CO ₂ sequestration	0.8x
Indirect (ecosystem)	Watershed services	0.6x
Indirect (community)	Livelihood improvement	0.4x
Systemic (policy)	Protected-area designation	0.5x
Total leverage		3.3x

Table 9: Conservation leverage decomposition.

7.2 Fund Lifecycle

1. **Issuance:** Donations arrive via DonationRouter. ImpactVouchers are minted.
2. **Allocation:** Treasury allocates across programs using impact-weighted weights.
3. **Disbursement:** Program contracts receive funds via TreasuryMultisig with tier-appropriate approval.
4. **Execution:** Conservation activities occur in the physical world.
5. **Measurement:** ImpactOracle publishes verified metrics.
6. **Settlement:** ImpactVouchers are settled: Verified, Partial, or Expired.
7. **Reporting:** Quarterly reports are auto-generated and published to IPFS.

This lifecycle is described in full in the companion platform paper [18].

8 Leverage Analysis

8.1 Direct and Indirect Impact

Conservation spending generates multiplier effects beyond the direct expenditure:

Definition 8.1 (Conservation Leverage Ratio).

$$\Lambda = \frac{I_{direct} + I_{indirect} + I_{systemic}}{C} \quad (1)$$

where I_{direct} is the value of direct conservation outcomes (species protected, habitat preserved), $I_{indirect}$ is the value of co-benefits (carbon sequestration, ecosystem services, community livelihoods), $I_{systemic}$ is the value of systemic changes (policy adoption, behavior change, knowledge creation), and C is the conservation expenditure.

The 3.3x leverage ratio means each dollar of Zoo conservation spending generates \$3.30 in total conservation value when indirect and systemic effects are included. This is consistent with meta-analyses of conservation ROI [8, 9].

Risk	Mitigation	Residual
Program failure	Adaptive management, portfolio diversification	Medium
Climate impacts	Climate-resilient site selection	Medium
Political instability	Multi-country presence	Low
Funding gaps	Diversified: donations + validator splits	Low
Mission drift	Immutable conservation floors + ethics board	Low
Greenwashing	Independent oracle verification	Low

Table 10: Risk matrix.

8.2 Network Effect

Zoo’s on-chain infrastructure creates an additional leverage mechanism: as the network grows, the ConservationFeeSplitter generates more passive conservation funding without additional donor action. At 10x current network usage, the annual passive conservation funding would reach \$8.4M—equivalent to a medium-sized conservation NGO’s annual budget, generated entirely from transaction fees.

9 Risk Management

9.1 Conservation Risks

9.2 Community Risks

FPIC failure is the highest-consequence community risk. Mitigation: every community engagement follows a documented FPIC process with community veto power. FPIC status is tracked on-chain (IMP-008 sub-metric) and reported in quarterly transparency reports.

9.3 Technology Risks

Smart-contract vulnerability is the primary technology risk. Mitigation: all treasury and fund-management contracts require two independent security audits before mainnet deployment. The foundation maintains a \$1M bug bounty program funded from the Emergency Fund.

10 Alignment with Global Frameworks

10.1 Conservation Frameworks

- **IUCN Red List:** Species prioritization for monitoring and protection programs.
- **Key Biodiversity Areas (KBAs):** Site selection for habitat protection programs.
- **30x30 Target:** Area-based conservation contributing to the GBF target of 30% land and ocean protection by 2030.
- **Kunming-Montreal GBF:** Alignment with 5 of 23 GBF targets (see Table 4).

Dimension	Traditional Conservation	Zoo Approach
Ethics	Human interests paramount	All life has value
Communities	Beneficiaries	Rights-holders
Knowledge	Extracted	Protected, co-authored
Measurement	Self-reported, annual	Oracle-verified, continuous
Treasury	Opaque, audited annually	On-chain, real-time
Funding	Donation-dependent	Donation + protocol revenue
Scale	Linear with fundraising	Exponential with network growth
Permanence	Policy-dependent	Protocol-level invariants

Table 11: Structural differences between traditional and Zoo conservation.

10.2 Impact Measurement Standards

- **IRIS+**: Metric alignment for impact investing comparability (Table 3).
- **CARE Principles**: Indigenous data governance and sovereignty.
- **FAIR Principles**: Findable, Accessible, Interoperable, Reusable data for research outputs.
- **UN SDGs**: Contribution to SDG 14 (Life Below Water) and SDG 15 (Life on Land).

10.3 ESG Integration

The foundation’s impact data is structured for ESG reporting consumption [15]. Institutional investors seeking biodiversity-impact exposure can reference Zoo’s ImpactOracle metrics as part of their ESG disclosures, creating a bridge between on-chain conservation data and traditional financial reporting.

11 What Makes This Different

The most consequential difference is permanence. Traditional conservation funding depends on the continued goodwill of boards, donors, and governments. Zoo’s conservation funding is permanent at the protocol level: the immutable 15% floor in the ConservationFeeSplitter ensures that conservation funding persists as long as the network processes transactions.

12 Related Work

The Nature Conservancy operates a \$7.4B endowment for conservation but relies on traditional investment management and annual reporting [10]. WWF manages \$350M+ in annual conservation spending but acknowledged in 2022 that monitoring and evaluation remain the weakest link [5]. The Ocean Foundation pioneered blue economy finance but operates off-chain [11]. GiveWell’s cost-effectiveness analysis provides the gold standard for impact evaluation but applies ex-post with multi-year lag [6]. Climate Bonds Initiative has developed certification for green bonds but focuses on climate rather than biodiversity [12]. Regen Network issues ecological credits on-chain but targets carbon markets [13]. Impact-linked finance instruments (Development Impact Bonds, Social Impact Bonds) share the outcome-based philosophy but operate on traditional banking rails [14].

13 Future Work

1. **Biodiversity credits:** Develop a standardized on-chain biodiversity credit instrument based on ImpactOracle metrics, enabling market-based conservation finance.
2. **Insurance pool:** Create a mutual insurance pool for conservation programs, funded from the Emergency Fund and validator-reward splits.
3. **AI-driven allocation:** Use reinforcement learning to optimize impact-weighted allocation, maximizing conservation outcomes per dollar spent.
4. **Academic partnerships:** Establish data-sharing agreements with university conservation departments for peer-reviewed validation of ImpactOracle metrics.
5. **Legal framework:** Develop model legislation for on-chain nonprofit treasury management, in collaboration with the Uniform Law Commission.

14 Conclusion

Zoo Labs Foundation’s impact thesis is that conservation finance can be measurable, verifiable, and scalable through on-chain infrastructure. The combination of smart-contract fund management, oracle-verified impact metrics, protocol-level conservation funding, and a sub-DAO umbrella structure creates an organization that operates simultaneously as nonprofit, technology platform, and impact authority.

The thesis is testable. The targets are quantified. The metrics are standardized. The verification is independent. If the thesis is wrong—if on-chain conservation infrastructure does not produce better outcomes than traditional approaches—the on-chain data will show it. That transparency is itself the thesis.

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References

- [1] H. Paulson et al., “Financing Nature: Closing the Global Biodiversity Financing Gap,” The Paulson Institute, 2020.
- [2] Convention on Biological Diversity, “Kunming-Montreal Global Biodiversity Framework,” CBD/COP/15/L.25, 2022.
- [3] Global Impact Investing Network, “IRIS+ System: Impact Measurement Standards,” GIIN, 2023.
- [4] IUCN, “The IUCN Red List of Threatened Species,” Version 2023-1, 2023.
- [5] World Wildlife Fund, “Living Planet Report 2022,” WWF International, 2022.

- [6] GiveWell, “Cost-Effectiveness Analysis of Top Charities,” GiveWell Research, 2023.
- [7] Charity Navigator, “Giving Statistics: Charity Navigator’s 2023 Report,” Charity Navigator, 2023.
- [8] A. Waldron et al., “Reductions in Global Biodiversity Loss Predicted from Conservation Spending,” *Nature*, vol. 551, pp. 364–367, 2019.
- [9] A. Balmford et al., “Global Variation in Terrestrial Conservation Costs,” *Biological Conservation*, vol. 117, pp. 225–232, 2003.
- [10] The Nature Conservancy, “2023 Annual Report,” The Nature Conservancy, 2023.
- [11] The Ocean Foundation, “Blue Economy and Ocean Conservation Finance,” Technical Report, 2021.
- [12] Climate Bonds Initiative, “Green Bond Market Summary 2023,” Climate Bonds Initiative, 2023.
- [13] Regen Network, “Regen Ledger: Ecological State Protocols,” Regen Network Whitepaper, 2021.
- [14] E. Gustafsson-Wright et al., “Impact Bonds in Developing Countries,” Brookings Institution, 2019.
- [15] CFA Institute, “ESG Integration in Practice,” CFA Institute Research, 2022.
- [16] IPBES, “Global Assessment Report on Biodiversity and Ecosystem Services,” IPBES Secretariat, 2019.
- [17] OECD, “Biodiversity and the Economic Response to COVID-19,” OECD Policy Responses, 2020.
- [18] A. Worrying and Z. Kelling, “Zoo Fund: A Transparent Conservation Finance Platform,” Zoo Labs Foundation, 2024.
- [19] A. Worrying and Z. Kelling, “Impact Metric Oracles: Verifiable Conservation Outcomes On-Chain,” Zoo Labs Foundation, 2025.
- [20] A. Worrying and Z. Kelling, “Decentralized Treasury Management for a Conservation NGO,” Zoo Labs Foundation, 2025.
- [21] A. Worrying and Z. Kelling, “Conservation-Aligned Validator Rewards,” Zoo Labs Foundation, 2025.
- [22] A. Worrying and Z. Kelling, “Biodiversity Impact Bonds: Tokenized Conservation Finance,” Zoo Labs Foundation, 2025.
- [23] Gnosis, “Safe: Multi-Signature Wallet Infrastructure,” Technical Documentation, 2021.
- [24] Chainlink Labs, “Decentralized Oracle Networks: Architecture and Security,” Chainlink Whitepaper v2, 2023.
- [25] Fundraising Effectiveness Project, “2024 Quarterly Fundraising Report,” AFP/GivingTuesday, 2024.

- [26] United Nations, “Transforming Our World: The 2030 Agenda for Sustainable Development,” UN General Assembly, 2015.
- [27] United Nations, “Declaration on the Rights of Indigenous Peoples,” UN General Assembly, 2007.